

EEE 416 - Microprocessor and Embedded Systems Laboratory

January 2026 Level - 4 Term - 1 Section B1

Final Project Demonstration

PRIVACY-PRESERVING HUMAN PRESENCE & Motion DETECTION USING WI-FI CSI

SUBMITTED BY – GROUP B1-06



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Outline

1. Summary
2. Introduction
3. Design
4. Implementation
5. Analysis and Evaluation
6. References



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Wi-Fi CSI Human Presence & Activity Sensing

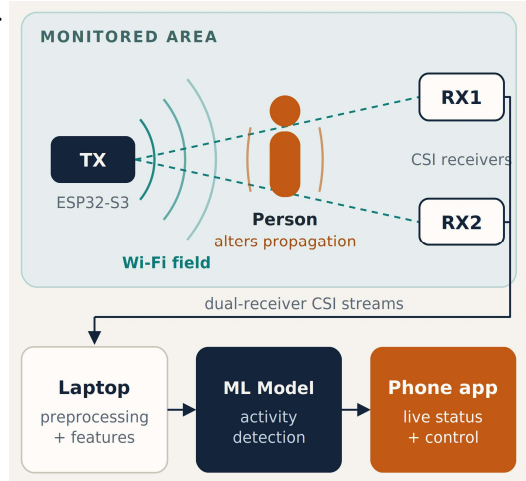
Presented by: 2106083 2

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Summary / Abstract

Objective: To develop a camera-free human sensing system using Wi-Fi Channel State Information (CSI) captured by ESP32-S3 devices.

- 1 ESP32-S3 transmitter + 2 CSI receivers
- CSI preprocessing and feature extraction
- ML-based presence and movement detection
- Automatic empty-room calibration
- Real-time phone monitoring and control
- Privacy-preserving: no camera, microphone, or wearable



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Introduction

CONVENTIONAL SENSING & ITS LIMITS

	CCTV / cameras	Raise privacy concerns
	PIR sensors	Mainly detect motion; may miss stationary occupants
	Wearables	Users must carry or wear a device
	Ultrasonic & others	Need extra dedicated hardware and infrastructure

Our project uses existing Wi-Fi signals and Channel State Information (CSI) to detect room occupancy and gross human movement without cameras, wearables, or additional sensing infrastructure.



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Why ESP32-S3 Instead of a Usual Router?

Conventional Wi-Fi router



- ✗ Stock firmware does not expose CSI to the user
- ✗ CSI access needs special NICs or modified firmware
- ✗ Little control over packet rate, channel and timestamps
- ✗ Bulky, mains-powered and costly to deploy as many receivers

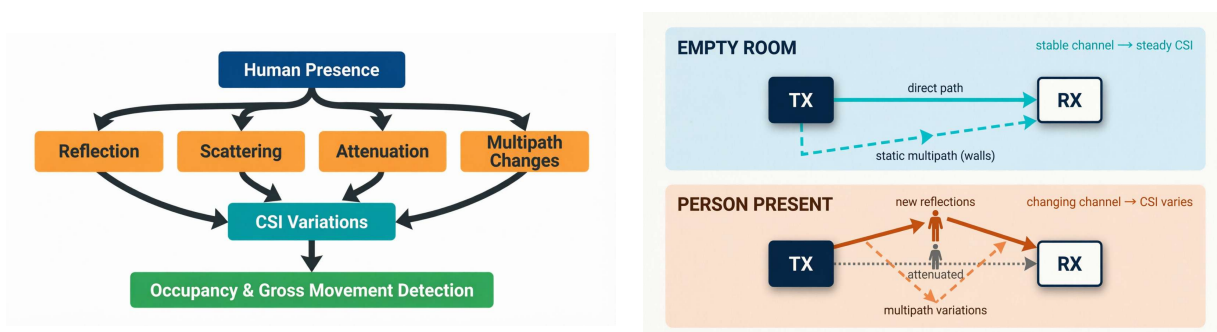
ESP32-S3 (our choice)



- ✓ Built-in CSI support in Espressif's ESP-IDF
- ✓ Low cost: only a few Taka per board
- ✓ Small and USB-powered: easy to place 1 TX + 2 RX
- ✓ Full control of packets and timing; streams CSI over serial

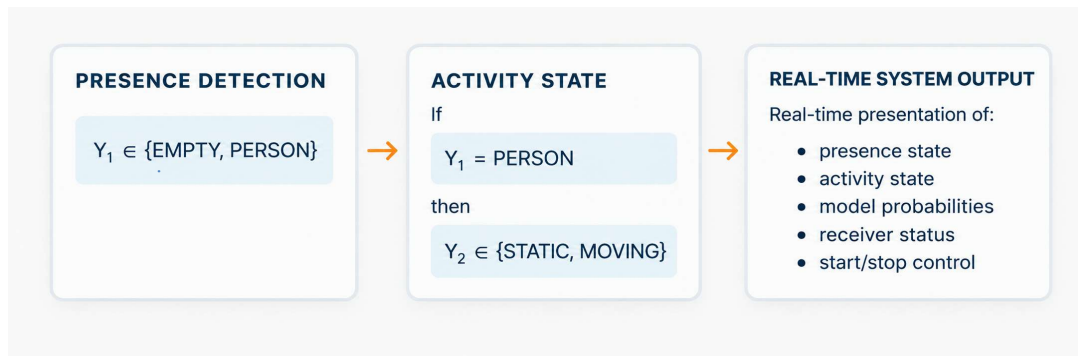
How Human Presence Changes the Wi-Fi Channel

Channel State Information (CSI) is information that describes how a Wi-Fi signal changes while traveling from the transmitter to the receiver.



Design: Problem Formulation (PO(b))

Identification of Scope



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Design: Problem Formulation (PO(b))

Literature Review

Existing sensing families

Approach	Advantage	Limitation
Camera vision	Rich information	Privacy + lighting
PIR	Cheap/simple	Mostly motion based
Wearables	Good individual information	Must be worn
Radar	Strong sensing capability	Dedicated RF hardware
Wi-Fi CSI	Device-free and privacy-friendly	Environment dependent

Our direction

Rather than reconstructing images or human pose, this project uses:
engineered CSI statistics + lightweight classical ML



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Design: Problem Formulation (PO(b))

Formulation of Problem

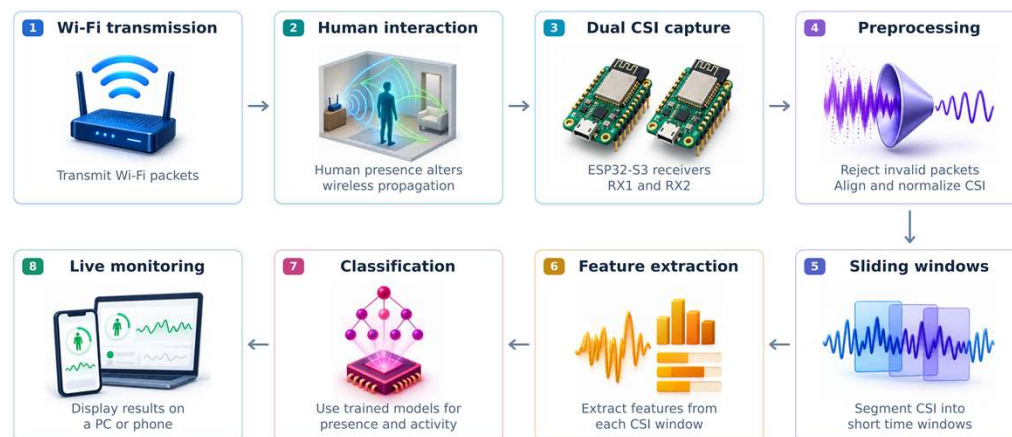
Requirements

- ✓ capture synchronized dual-receiver CSI continuously
- ✓ reject malformed packets
- ✓ align both receivers on a common time base
- ✓ compensate for common amplitude scaling
- ✓ detect environmental deviation from an EMPTY baseline
- ✓ distinguish static and moving human presence
- ✓ operate live
- ✓ expose results to a phone



Design: Problem Formulation (PO(b))

Wi-Fi CSI-Based Detection Workflow:



Design: Problem Formulation (PO(b))

| Wi-Fi Transmission



Wi-Fi packets probe the room.

The transmitter sends a continuous stream of wireless packets toward the receivers.

OUTPUT

Wireless signals in the sensing area



Design: Problem Formulation (PO(b))

| Human Interaction



Human presence changes the wireless channel.

People reflect, block and scatter Wi-Fi signals, changing the paths that reach each receiver.

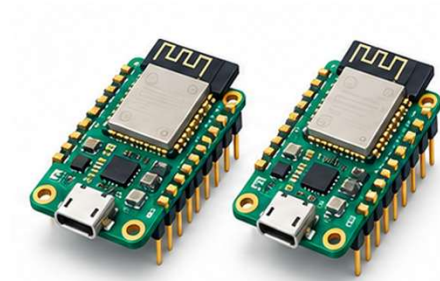
OUTPUT

Changes in the received channel



Design: Problem Formulation (PO(b))

Dual CSI Capture



Two receivers collect CSI.

ESP32-S3 RX1 and RX2 capture channel measurements from received packets and stream them to the PC.

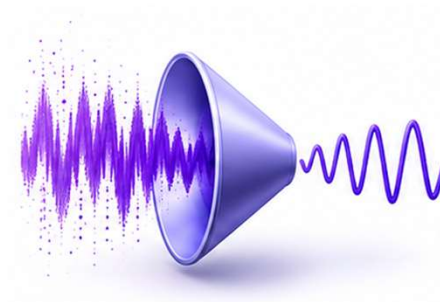
OUTPUT

Two CSI streams



Design: Problem Formulation (PO(b))

Preprocessing



Prepare reliable CSI samples.

Reject malformed packets, align both streams on a common time base and compensate for amplitude scaling.

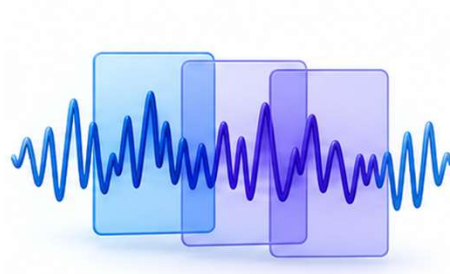
OUTPUT

Aligned and normalized CSI



Design: Problem Formulation (PO(b))

| Sliding Windows



Analyze short sections of CSI.

A sliding window groups recent samples. New samples move the window forward for the next decision.

OUTPUT

Successive CSI windows



Design: Problem Formulation (PO(b))

| Feature Extraction



Describe channel changes with features.

Summarize CSI variation and deviation from the EMPTY baseline in each window.

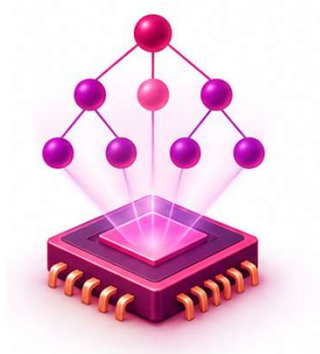
OUTPUT

A feature vector for classification



Design: Problem Formulation (PO(b))

Classification



Decide presence, then activity.

First classify EMPTY or PERSON.
When a person is present, classify
STATIC or MOVING.

OUTPUT

Presence and activity labels



Design: Problem Formulation (PO(b))

Live Monitoring



Show live results on PC and phone.

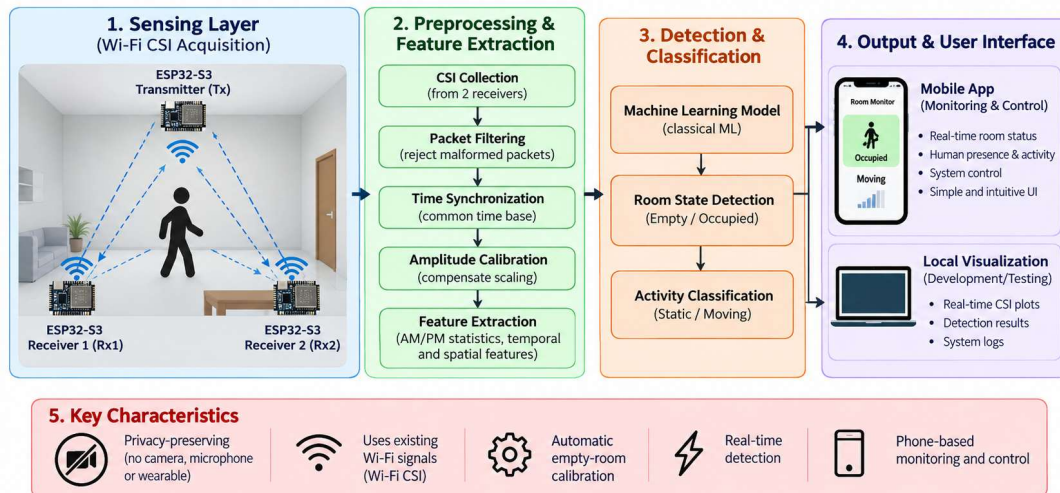
Display the detected state,
confidence and receiver status.
Provide controls to start or stop
monitoring.

OUTPUT

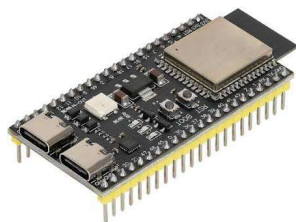
A live view of presence and activity



Design: Overall System Architecture



Hardware Components



ESP32-S3



USB Cable



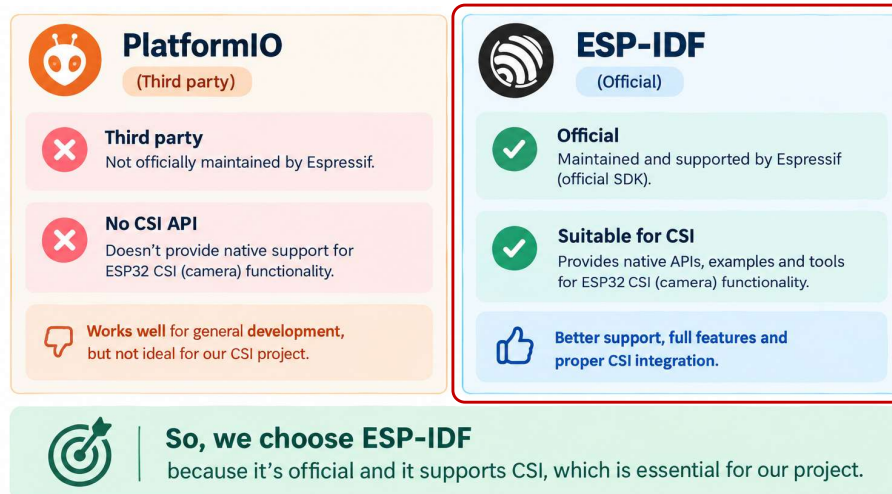
Power bank

Why choose ESP32 s3



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Software Choice



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Firmware Setup

Workflow of ESP-IDF 5.5.5 in VS Code



Firmware Setup (Cont)

After Setup of TX

```

I (561) WIFI_AP: =====
I (561) WIFI_AP: WiFi Access Point Started
I (571) WIFI_AP: SSID      : fahim
I (571) WIFI_AP: Password  : (hidden)
I (571) WIFI_AP: Channel   : 6
I (581) WIFI_AP: Max STA   : 8
I (581) WIFI_AP: Bandwidth : 20 MHz
I (581) WIFI_AP: =====
I (3591) UDP_TX: UDP Sender Started
I (3591) UDP_TX: Destination : 192.168.4.255
I (3591) UDP_TX: Port        : 3333
I (3591) UDP_TX: Packet Size : 256 Bytes
I (3591) UDP_TX: Interval    : 5 ms
  
```

1 TX starts the access point

After Setup of RX

```

I (78179) wifi:state: init -> auth (0xb0)
I (78189) wifi:state: auth -> assoc (0x0)
I (78199) wifi:state: assoc -> run (0x10)
I (78269) wifi:connected with fahim, aid = 1, channel 6, BW20, bssid = e8:3d:c1:f1
199:69
I (78269) wifi:security: WPA2-PSK, phy: bgn, rssi: -21, cipher(pairwise:0x3, group
:0x3), pmf:1
I (78279) wifi:pm start, type: 1
I (78279) wifi:set rx beacon pti, rx_bcn_pti: 0, bcn_timeout: 25000, mt_pti: 0, mt
_time: 10000
I (78339) wifi:AP's beacon interval = 102400 us, DTIM period = 1
I (79389) esp_netif_handlers: sta ip: 192.168.4.2, mask: 255.255.255.0, gw: 192.16
8.4.1
I (79389) WIFI_STA: =====
I (79389) WIFI_STA: Connected Successfully!
I (79389) WIFI_STA: IP Address : 192.168.4.2
I (79399) WIFI_STA: =====
  
```

2 RX joins as a station



Challenges faced during setup of Tx and RX

```

I (561) WIFI_AP: =====
I (561) WIFI_AP: WiFi Access Point Started
I (571) WIFI_AP: SSID      : fahim
I (571) WIFI_AP: Password : (hidden)
I (571) WIFI_AP: Channel  : 6
I (581) WIFI_AP: Max STA  : 8
I (581) WIFI_AP: Bandwidth: 20 MHz
I (581) WIFI_AP: =====
I (3591) UDP_TX: UDP Sender Started
I (3591) UDP_TX: Destination : 192.168.4.255
I (3591) UDP_TX: Port       : 3333
I (3591) UDP_TX: Packet Size : 256 Bytes
I (3591) UDP_TX: Interval   : 5 ms
  
```

1 TX starts the access point

Problem 1: Broadcast

Interference with other Wi-Fi signals
Too poor CSI signal strength

Problem 2: Packet Rate very high

```

W (78919) WIFI_STA: Disconnected. Reconnecting...
W (73329) WIFI_STA: Disconnected. Reconnecting...
W (75739) WIFI_STA: Disconnected. Reconnecting...
W (78149) WIFI_STA: Disconnected. Reconnecting...
  
```

- 1 TX sent a packet every 5 ms → 200 CSI lines per second per RX
- 2 That is more text than the USB serial link can carry
- 3 Printing blocked the CSI callback, starving the Wi-Fi driver
- 4 The RX dropped off the network: "Disconnected. Reconnecting..."



Solution

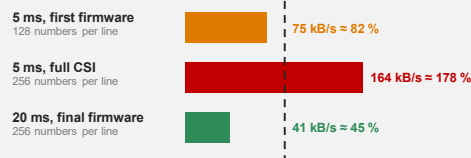
```

I (610) WIFI_AP: =====
I (610) WIFI_AP: WiFi Access Point Started
I (620) WIFI_AP: SSID      : fahim
I (620) WIFI_AP: Password : (hidden)
I (620) WIFI_AP: Channel  : 6
I (630) WIFI_AP: Max STA  : 8
I (630) WIFI_AP: Bandwidth: 20 MHz
I (630) WIFI_AP: =====
I (3640) UDP_TX: UDP Sender Started (Dual Unicast)
I (3640) UDP_TX: Targets: 192.168.4.2, 192.168.4.3 Port: 3333 | Interval: 20 ms
  
```

What we changed

- Interval 5 ms → 20 ms
- Broadcast → unicast

Text each RX must send over USB (kB/s)



Firmware Setup (Cont)

```
I (10320) wifiip: 1, bi: 102400, li: scale list interval from 307200 us to 307200 us
I (10320) wifi:set rx beacon pti, rx_bcn_prti: 0, bcn_timeout: 25000, mt_ptti: 0, mt_time: 10000
I (10330) esp_netif_handlers: sta ip: 192.168.4.2, mask: 255.255.255.0, gw: 192.168.4.1
I (10340) WIFI_STA:
I (10340) WIFI_STA: Connected Successfully!
I (10350) WIFI_STA: IP Address : 192.168.4.2
I (10350) WIFI_STA: =====
I (10360) CSI: TX/AP BSSID: 28:37:2f:c1:7f:4d
I (10360) CSI: CSI configuration accepted
I (10370) CSI: CSI enabled successfully
I (10370) wifiAP's beacon interval = 102400 us, DTIM period = 1
I (10380) main task: Returned from ap_main()
I (12000) wifi:cb4-addbdc0 (ifx0, 28:37:2f:c1:7f:4d), tid0, sn0, ssn0, vsm0=64
I (12000) 0x19020473, c1, 98,6,4,1,3,0,0,256,0,0 [0,0,9,8,5,8,5,8,3,8,3,9,3,8,1,8,1,8,1,8,0,9,0,9,-1,8,1,9,0,9,1,8,0,8,1,6,1,
8,1,7,7,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,12,7,11,9,10,10,10,10,11,9,12,9,12,8,13,7,13,14,7,14,6,13,7,13,16,11,6,12,6,12,7,7,11,7,11,8,11
6,7,10,7,10,8,18,10,9,7,8,9,9,0,0,4,-50,1,-52,0,-51,-6,42,-8,43,-6,46,-10,-40,-7,-38,-12,32,9,-43,-12,-42,-17,-40,-18,-35,-19,-34,-21,-
36,-22,-35,-20,-34,-20,-35,-21,-35,-19,-32,-22,-36,-18,-33,-21,-32,-19,-33,-16,-33,-14,-34,-9,-29,-10,-34,0,0,0,0,0,0,0,0,0,0,-8,-53,-5,-5
4,-1,-56,4,-59,10,-62,12,-65,19,-56,18,-58,22,-66,23,-55,25,-60,28,-54,28,-54,29,-50,24,-45,22,-50,19,-49,21,-53,18,-53,14,-42,11,-47,13,-54,6,-47,
9,-51,6,-49,1,-50,0,-40,-49]
```

RX1 Ready

RX2 Ready

[illegible]

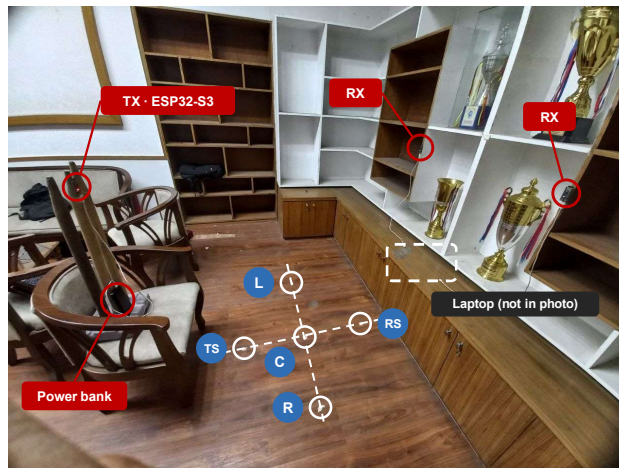
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Dataset Collection: The Sensing Area

Everything in one view: transmitter, receivers, laptop position and the five marked spots



What is where

TX

RX

Laptop

TS C RS

LR

...

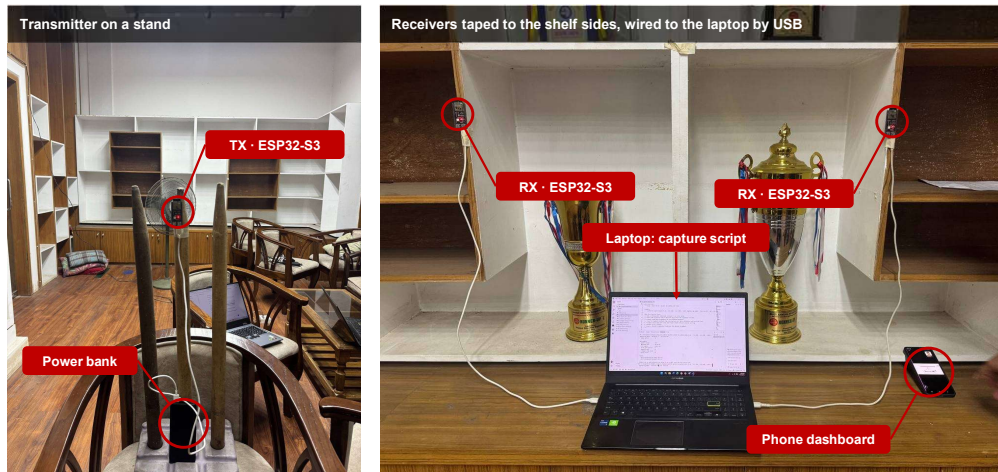


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Wi-Fi CSI Human Presence & Activity Sensing Presented by: 2106078 28

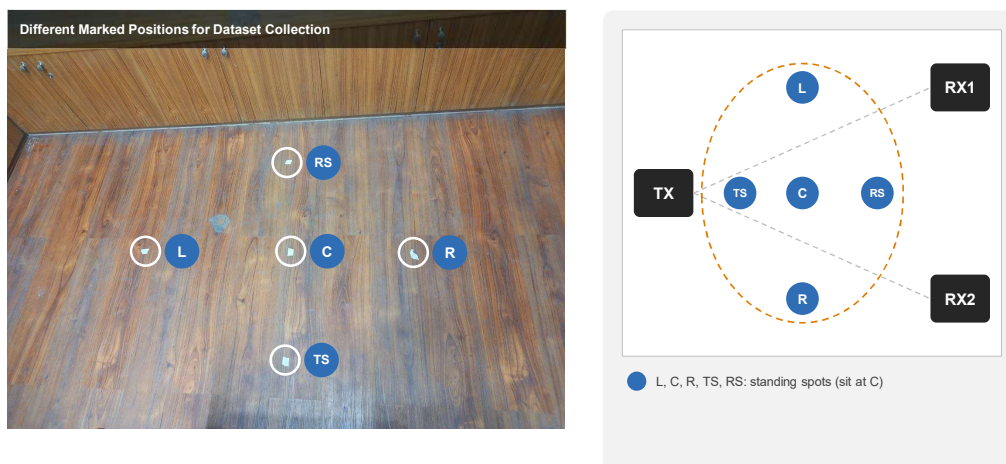
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Hardware & Firmware: Setup in the Room



Dataset Collection: Marked Positions

Paper markers on the floor put every subject at exactly the same spots in every session



Dataset Collection: Recording Procedure

Each session: one subject, one position or path, 60 s, both receivers recorded together



Recording settings

- 60s per session
- 5-s countdown
- TX and receivers never moved between sessions
- One session saves two .csv files (RX1, RX2)



Dataset Collection: List of Datasets

39 sessions × 2 receivers = 78 raw CSV files, each about 60 s long

Scenario (file label)	Type	S01	S02	S03
empty_01 ... empty_09	EMPTY	9 sessions, no subject		
stand_center	STATIC	✓	✓	✓
stand_left	STATIC	✓	✓	✓
stand_right	STATIC	✓	✓	✓
sit_center	STATIC	✓	✓	✓
stand_txside	STATIC	✓	✓	✓
stand_rxside	STATIC	✓	✓	✓
normal_walk	MOVING	✓	✓	✓
slow_walk	MOVING	✓	✓	✓
walk_left_right	MOVING	✓	✓	✓
walk_tx_rx	MOVING	✓	✓	✓

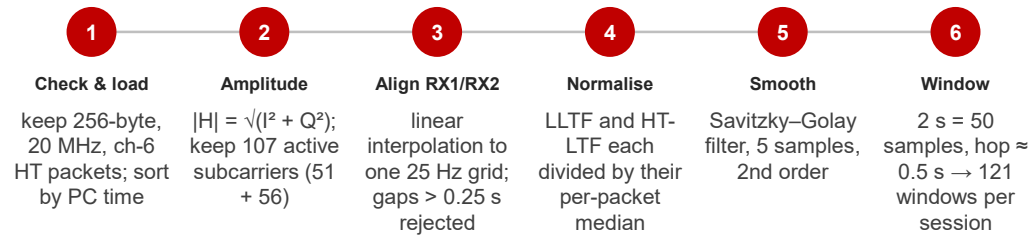
**39
Sessions**

**78 CSV
Files**



Data Preprocessing: Pipeline

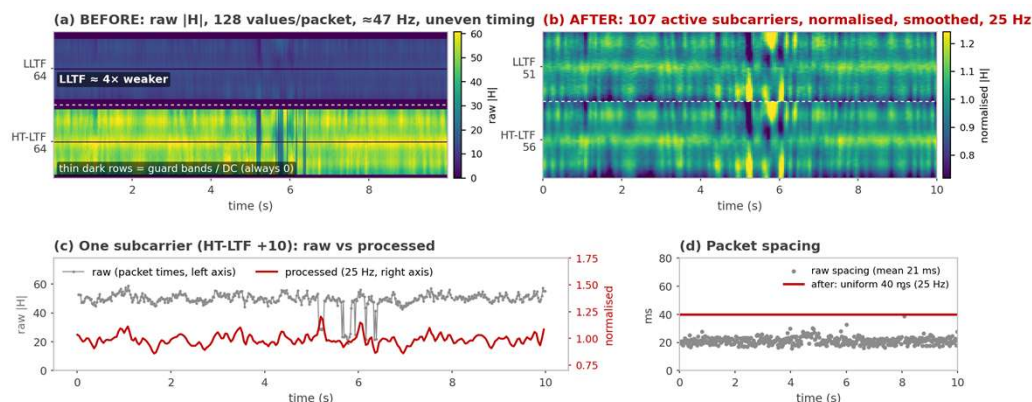
From raw CSV rows to clean 2-s windows — the same code builds the dataset and runs live



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Data Preprocessing: Before vs After

Real RX1 recording: subject S01 walking TX $\leftrightarrow$ RX, seconds 20–30



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Data Preprocessing: Processed Datasets

What the pipeline writes to data/processed (Phase 1 folder shown)

all_features	9/23/2026 12:56 AM	Microsoft Excel Co...	1,418 KB
empty_baseline.npz	9/23/2026 12:54 AM	NPZ File	2 KB
metadata	9/23/2026 12:56 AM	JSON Source File	2 KB
session_summary	9/23/2026 12:56 AM	Microsoft Excel Co...	1 KB
test_features	9/23/2026 12:56 AM	Microsoft Excel Co...	426 KB
train_features	9/23/2026 12:56 AM	Microsoft Excel Co...	569 KB
val_features	9/23/2026 12:56 AM	Microsoft Excel Co...	425 KB

3,630

windows = 30
sessions × 121

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features per window
(Phase 1)

by subject

train S01
val S02
test S03

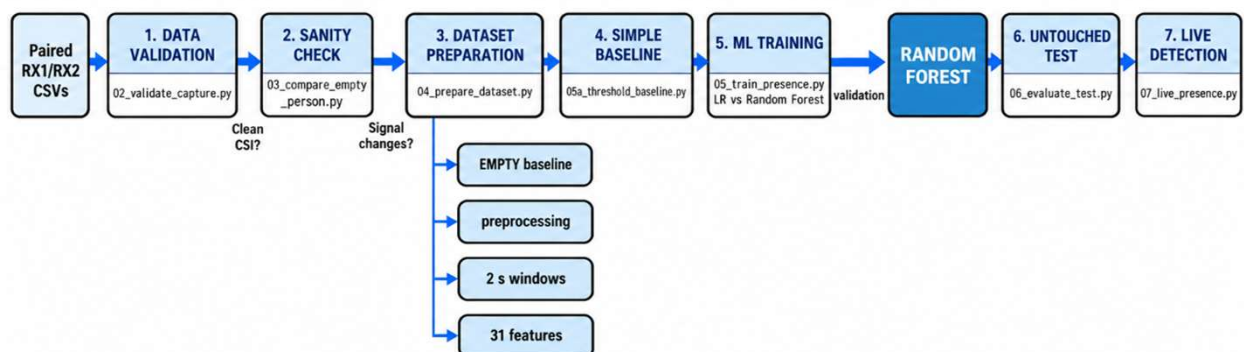
3,993 × 75

Phase 2 table:
windows × features



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Implementation: Phase 1 (Person/Empty)



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02_validate_capture.py: Data Quality Check

File	Rows	Duration	Rate	Seq miss	CSI len
Empty RX1	2648	59.96 s	44.15 Hz	0	256
Empty RX2	2394	59.95 s	39.91 Hz	0	256
Person RX1	2770	59.96 s	46.18 Hz	0	256
Person RX2	2655	59.95 s	44.27 Hz	0	256

Output of this stage

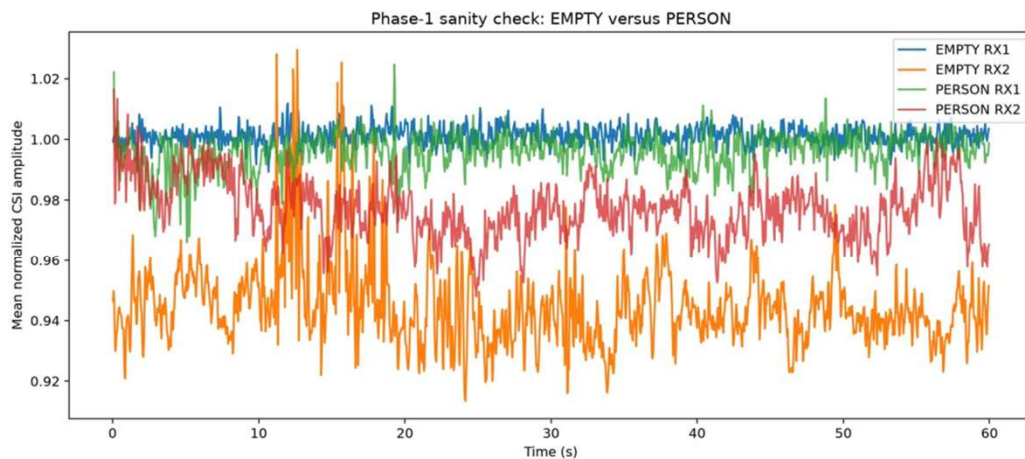
Accepted sessions move forward; recordings with gaps, missing packets, wrong CSI length or parsing errors are recollected or rejected.

Actual accepted examples showed zero sequence loss and CSI length 256.



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03_compare_empty_person.py: Sanity Check



RX1 RMSE

0.020924

RX2 RMSE

0.067379

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04_prepare_dataset.py + csi_utils.py: Preprocess, Window, Feature Extract

Split	Sessions	Windows	EMPTY	PERSON
Train	12	1452	605	847
Val	9	1089	242	847
Test	9	1089	242	847

Output: 31 features per 2-second window, using a 25 Hz standardized processing grid.



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05a_threshold_baseline.py: Simple Baseline Before ML

Decision rule

PERSON if
Dbaseline ≥ 0.022511673

Validation confusion matrix

	Pred E	Pred P
Actual E	238	4
Actual P	70	777

Accuracy
93.20%

Balanced Acc..
95.04%

Person Recall
91.74%

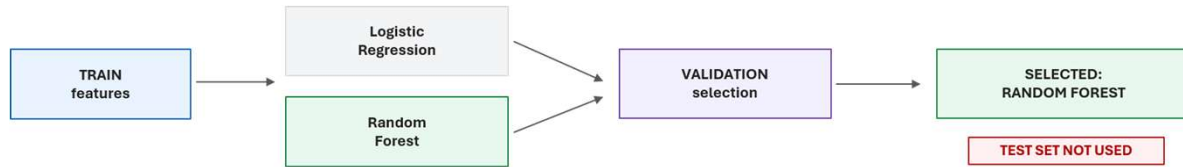
Macro-F1
91.00%

ROC-AUC
97.40%



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05_train_presence.py: Train and Select the Phase-1 Model

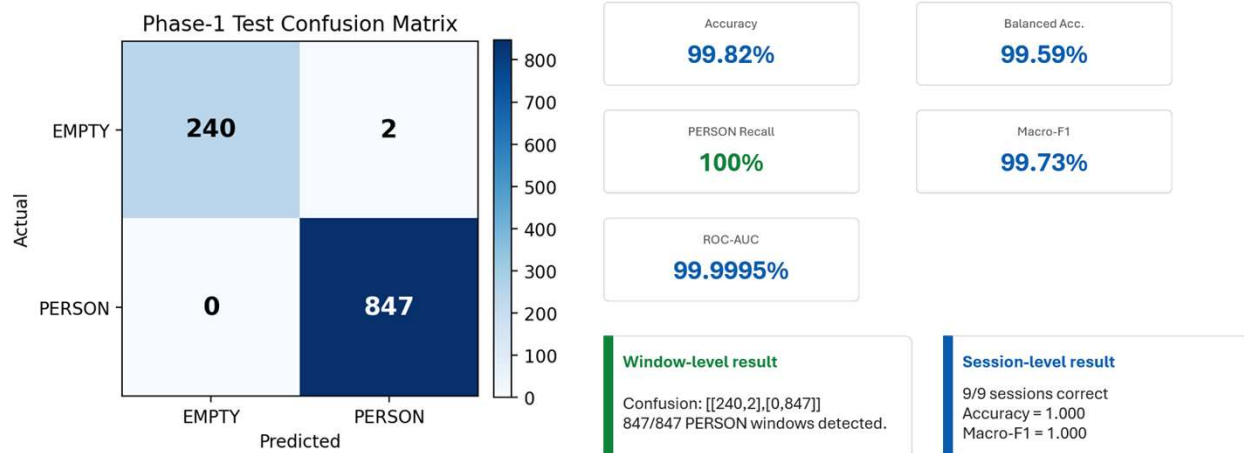


Model	Accuracy	Bal. Acc.	Macro-F1	ROC-AUC	Confusion
Logistic Regression	0.99725	0.99823	0.99603	0.99997	[[242,0],[3,844]]
Random Forest	0.99908	0.99941	0.99867	1.00000	[[242,0],[1,846]]



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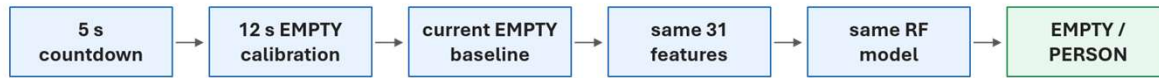
06_evaluate_test.py: Untouched Test Results



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07_live_presence.py: Offline Model to Live Detector

Final live setup



Live settings

25 Hz • 2.0 s window • 0.5 s update • 5-vote majority

What changed

Fresh current-room EMPTY baseline

What stayed fixed

Same trained Random Forest



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Phase 2: From Presence to Activity



EMPTY

no person

No one in the sensing area; the CSI stays close to the empty-room baseline.



STATIC

person, still

A person is present but still: standing (centre, left, right) or sitting.



MOVING

person, walking

A person is walking: normal, slow, left↔right or along the TX↔RX line.



Reused from Phase 1

- ESP32-S3 TX/RX firmware and the 50 Hz CSI link
- Dual-receiver capture script with a shared PC time base
- CSI preprocessing library (csi_utils.py)
- Presence model (Random Forest) and all EMPTY, STATIC and normal-walk recordings



New in Phase 2

- 9 new 60-s walking sessions (3 subjects × 3 walking styles)
- 44 baseline-free motion features per window
- STATIC/MOVING classifier, compared with a direct 3-class model
- Real-time detector with parallel Phase-1 / Phase-2 voting
- Phone dashboard to START, STOP and watch live results



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02_validate_phase2_new.py: Quality Check for New Moving Sessions

Session	RX1 rows	RX2 rows	RX1 Hz	RX2 Hz	Seq miss
S01 slow	2929	2974	48.84	49.60	0
S01 L/R	2961	2891	49.37	48.21	0
S01 TX/RX	2831	2848	47.21	47.51	0
S02 slow	2991	2979	49.86	49.66	0
S02 L/R	2701	2630	45.03	43.85	0
S02 TX/RX	2891	2890	48.21	48.19	0
S03 slow	2855	2831	47.61	47.21	0
S03 L/R	2648	2499	44.16	41.67	0
S03 TX/RX	2875	2887	47.95	48.15	0

PHASE-2 NEW MOVING DATA QC

S01_slow_walk RX1: PASS rows=2929 rate=48.84Hz seqmiss=0
 S02_walk_left_right RX2: PASS rows=2630 rate=43.85Hz seqmiss=0
 S03_walk_tx_rx RX2: PASS rows=2887 rate=48.15Hz seqmiss=0

Saved: ..\reports\new_data_qc.csv
 OVERALL: PASS

Min rate

41.67 Hz

Max rate

49.86 Hz

Seq miss

0



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03_prepare_phase2_dataset.py: Create Phase-2 Feature Dataset

Design size

33 sessions total
 9 EMPTY + 12 STATIC + 12
 MOVING

Feature size

31 base features
 +44 motion features
 =75 total features

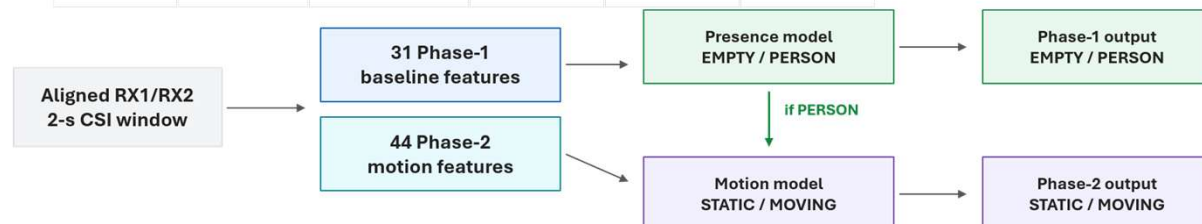
Windowing

2.0 s windows
 25 Hz processing
 121 windows/session

Subject split

S01 → train
 S02 → validation
 S03 → test

Split	Sessions	Windows	EMPTY	STATIC	MOVING
Train	13	1573	605	484	484
Val	10	1210	242	484	484
Test	10	1210	242	484	484



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04_train_phase2_models.py: STATIC vs MOVING Classifier

Model	Macro-F1	Bal. Acc.	Confusion matrix
Logistic Regression	0.945083	0.945248	[[484,0],[53,431]]
Random Forest	0.962787	0.962810	[[478,6],[30,454]]

LR validation

	Pred S	Pred M
Actual S	484	0
Actual M	53	431

RF validation

	Pred S	Pred M
Actual S	478	6
Actual M	30	454



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Phase 2: Results on Unseen Subject S03

Cascade — test confusion matrix

windows	Pred. EMPTY	Pred. STATIC	Pred. MOVING
True EMPTY	240	1	1
True STATIC	0	419	65
True MOVING	0	16	468

93.1%

window accuracy of the cascade

10 / 10

sessions correct by majority vote

99.2%

EMPTY windows recognised



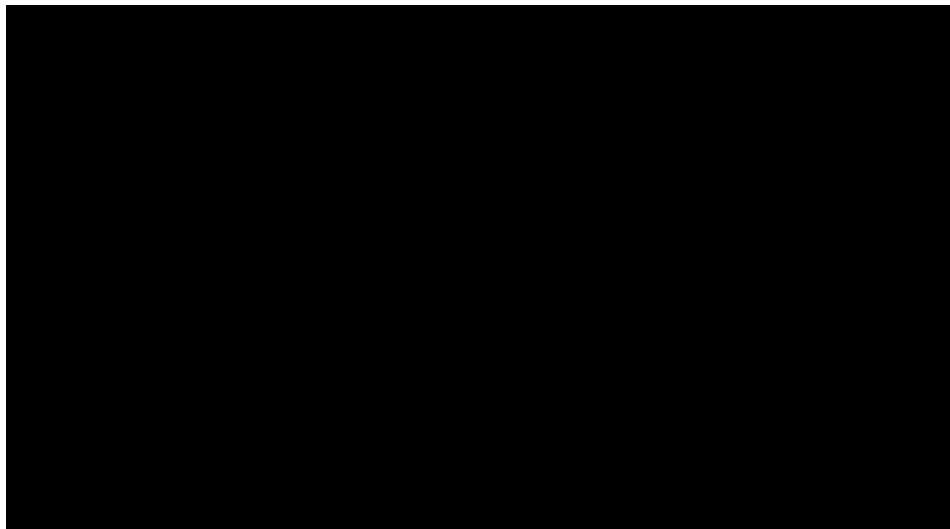
50

Implementation: Phone App



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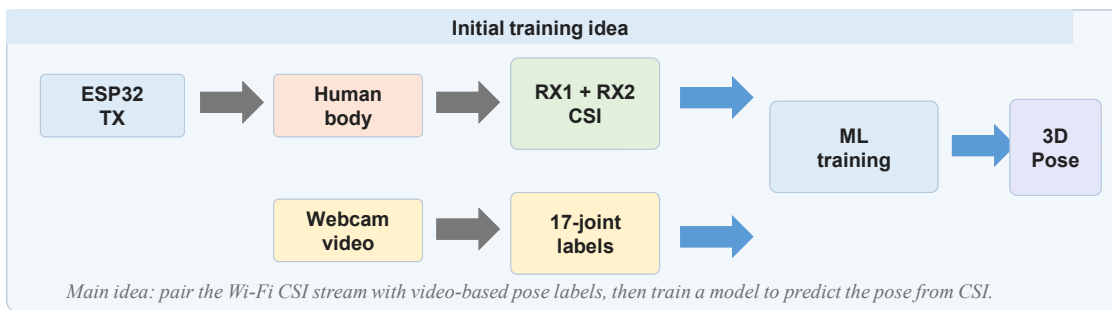
Video Demonstration



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Initial 3D Pose Estimation Attempt (Failed)

- ❑ Our first target was to estimate a 17-joint 3D human pose from Wi-Fi CSI.
- ❑ During training, video frames were used only to create pose labels.
- ❑ The main idea was that, after training, the model would use only the CSI data to predict the 3D human pose without using any camera.



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Video-Based Ground Truth Generation

Correct pose extraction examples

17-joint 3D skeleton

Live processing preview

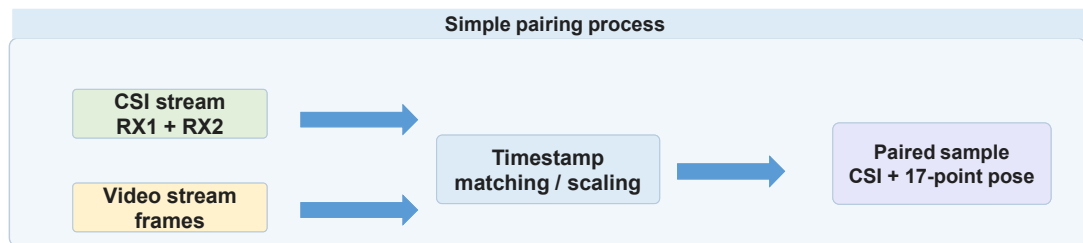
Live webcam preview

Example unreliable pose

- ❑ First, we used video frames to generate 17-joint pose labels.
- ❑ In many cases, the pose extraction worked well, but sometimes the joints were distorted or unstable.
- ❑ These pose labels were used as ground truth for training.

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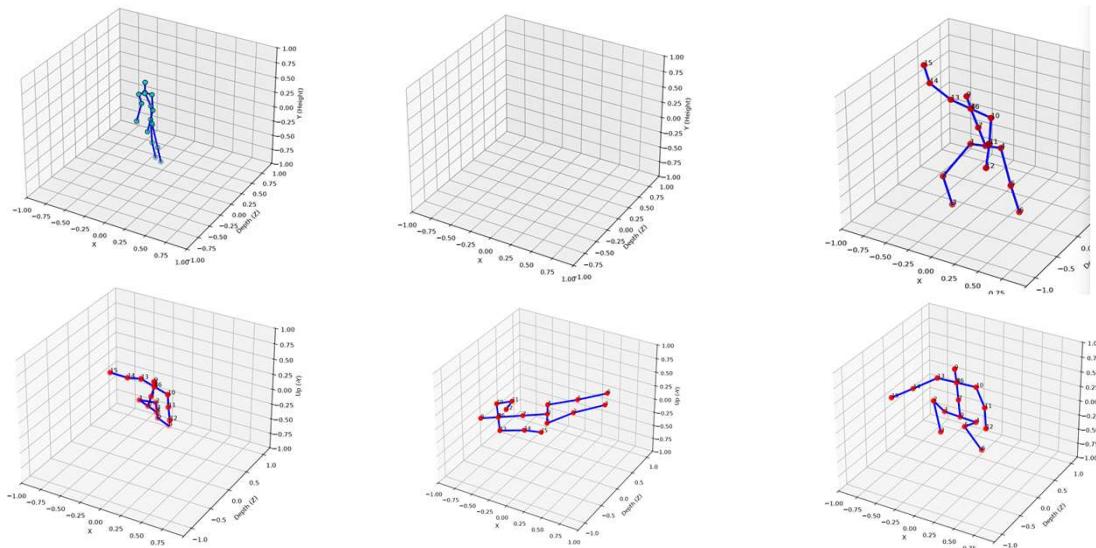
Dataset Creation: CSI and Video Were Matched



- ❑ We collected CSI data and video data at the same time.
- ❑ Then we matched their timestamps so that each CSI segment had the correct video pose label.
- ❑ In this way, we created paired CSI and pose data for training.
- ❑ The input was CSI data from two receivers, and the output was 17 body joints with X, Y and Z coordinates



Unpredictable Outputs



3D pose prediction was not stable and reliable enough for our project.

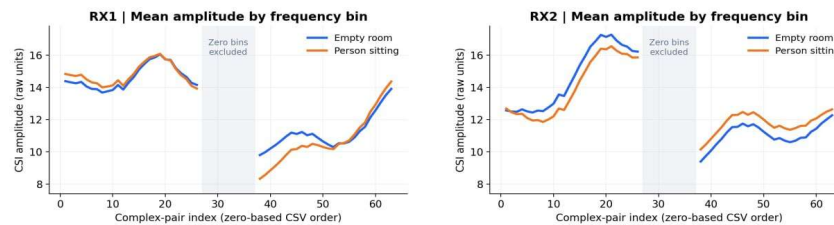


Why the 3D Pose Attempt Was Not Reliable

- ❑ The full CSI to 3D pose task was too difficult for the available data. Only 3 people were recorded, so the dataset was small.
- ❑ Some video-based pose labels were noisy or missing.
- ❑ CSI changed with the person, but the change was not enough to reconstruct a full 3D skeleton reliably.
- ❑ The model sometimes produced distorted joint positions.

Empty room vs. person sitting: CSI changes

Overall amplitude barely changes; the frequency pattern carries the clearer signal.



We also checked how CSI changed between an empty room and a person sitting. The signal did change, but in some parts the difference was very small. This showed that CSI was useful for sensing presence, but full 3D pose estimation was too difficult for our available data.



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5. Design Analysis and Evaluation



5.1 Novelty

What is new in our system



5.5 Impact Assessment (PO(f))

Societal, safety and legal impact



5.2 Design Considerations (PO(c))

Health and safety, environment, society



5.6 Sustainability Evaluation (PO(g))

Reuse, energy and efficiency



5.3 Investigations (PO(d))

Experiments, data and results



5.7 Ethical Issues (PO(h))

Privacy, consent and honest reporting



5.4 Limitations of Tools (PO(e))

What our tools cannot tell us



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5.1 Novelty

A complete, low-cost CSI sensing system, built and evaluated end to end



End-to-end prototype

ESP32-S3 CSI capture, laptop processing and a phone dashboard working together



Two receivers, one clock

RX1 and RX2 aligned on a shared PC time base (25 Hz grid)



Careful signal handling

Packet layout is checked; LLTF and HT-LTF are normalised separately



Self-calibrating and fast

12-s empty-room calibration at start-up; presence and motion votes run in parallel



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5.2 Design Considerations (PO(c))



Public health & safety

- Low-voltage USB boards, no mains wiring
- Cables and power bank secured, no trip hazards
- Output is advisory, not a certified safety signal



Environment

- Stores numbers, not continuous video
- Boards, cables and mounts are reusable
- Laptop and always-on Wi-Fi still use energy



Cultural & societal needs

- Camera-free: suits privacy-sensitive spaces
- Nothing to wear or charge
- Clear text and colour states; occupants must be informed



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5.3 Investigations (PO(d))

Design of experiment and data collection



Fixed setup

Same room and device placement; only person, position and activity change



39 sessions × 60 s

9 empty-room sessions + 3 subjects (S01–S03), both receivers



Subject-wise split

S01 train · S02 validation · S03 test (used only once)



Clean data

78 files, 227,505 packets, 41.7–50.0 Hz, no sequence gaps

Key results (unseen subject S03)

99.82%

Phase 1 accuracy
(EMPTY vs PERSON)

93.14%

Phase 2 cascade
accuracy (3 classes)

10 / 10

Phase 2 test sessions
correct (majority vote)



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5.4 Limitations of Tools (PO(e))



CSI depends on firmware

Fixed 256-byte layout; LTF merging and channel filtering already applied by the receiver



Approximate timing

Laptop timestamps include USB and OS delays; no radio-clock synchronisation



Limited motion detail

25 Hz sampling and 2-s windows give only about 0.5 Hz frequency resolution



Offline is not live

Live accuracy and detection delay were not measured separately



Scores, not guarantees

Model probabilities are not calibrated; 5-window voting adds delay



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5.5 Impact Assessment (PO(f))



Societal & cultural

- Privacy-friendly occupancy information (lighting, room use)
- Occupants must know what is measured and who sees it
- Never used to judge identity or productivity



Health & safety

- Low electrical risk: USB-powered boards
- Main risks: loose mounts, cables and obstacles
- Disconnected or stale receivers must be handled safely



Legal

- Check radio and personal-data rules before deployment
- No regulatory certification yet
- Dashboard needs a login and HTTPS beyond the lab



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5.6 Sustainability Evaluation (PO(g))

Reusable hardware, but the energy use still has to be measured



Strengths

- Repairable, reusable development boards
- Numeric data only, no video storage
- Compact state logs can cut storage further



Costs

- Three radios, USB links and a laptop always on
- Wi-Fi power saving off to keep timing steady
- Real power use not measured yet

$$E = P \times t$$

Saving 1 W all year saves 8.76 kWh



Next steps

- Smaller always-on host or on-device inference
- Lower sampling rate only after an accuracy test
- Recycle e-waste through proper channels



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5.7 Ethical Issues (PO(h))



Minimal output

Only occupancy and coarse motion; no identity, no images



Consent and data care

Voluntary, explained participation; limited access; subjects coded S01–S03



Video kept separate

Video was used only in the early pose attempt, never in the final camera-free system



Honest reporting

Validation and test kept separate; the failed pose attempt and weak session are reported openly



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Reflection on Individual and Team work

Individual Contribution of Each Member

Individual Contribution

Student ID	Main Responsibilities	Key Contributions
2106078	ESP32 firmware development and hardware setup	- Developed transmitter and receiver firmware using ESP-IDF (including CSI extraction and serial streaming) - Configured Wi-Fi settings and packet transmission - Hardware integration, device testing and troubleshooting
2106080	CSI data acquisition and preprocessing pipeline	- Implemented synchronized data capture from RX1 and RX2 - Data validation, quality control and session management - CSI parsing, alignment, resampling and normalization - Developed dataset preparation pipeline
2106082	Phase 1 (EMPTY/PERSON) model development and evaluation	- Feature engineering (31 features) and EMPTY baseline construction - Implemented threshold baseline, Logistic Regression and Random Forest models - Model training, validation, and untouched test evaluation - Developed live presence detection module
2106083	Phase 2 (STATIC/MOVING) and mobile application	- Designed and implemented motion feature extraction (44 features) - Trained and evaluated motion classification models - Integrated hierarchical Phase 1 → Phase 2 inference - Developed web server and mobile phone interface for real-time monitoring and control



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Reflection on Individual and Team work Logbook of Project

Week	Activities
Week 3	Project assignment and topic selection.
Week 4–5	Studied Wi-Fi CSI, ESP32-S3, and relevant research papers.
Week 6–7	Designed system architecture and completed hardware/firmware setup.
Week 8–9	Collected and prepared dual-receiver CSI data.
Week 10–11	Performed preprocessing and developed presence detection model.
Week 12–13	Developed activity classification, phone interface, and completed system testing.
Week 14	Prepared documentation, final demo, presentation, and submission.



Project Management and Cost Analysis (PO(k))

Bill of Materials

Item	Quantity	Unit Cost (BDT)	Subtotal (BDT)
ESP32-S3 Development Board	3	750	2,250
Data-capable USB Cable	3	150	450
Nonmetal Enclosure / Stand Allowance	3	250	750
Labels, Cable Ties and Spare Parts	1	300	300
Laptop / Host Computer	Reused	0	0
Smartphone for Monitoring	Reused	0	0
Power Bank for Transmitter	Reused	0	0
Total Estimated Prototype Cost			3,750 BDT



Project Management and Cost Analysis (PO(k))

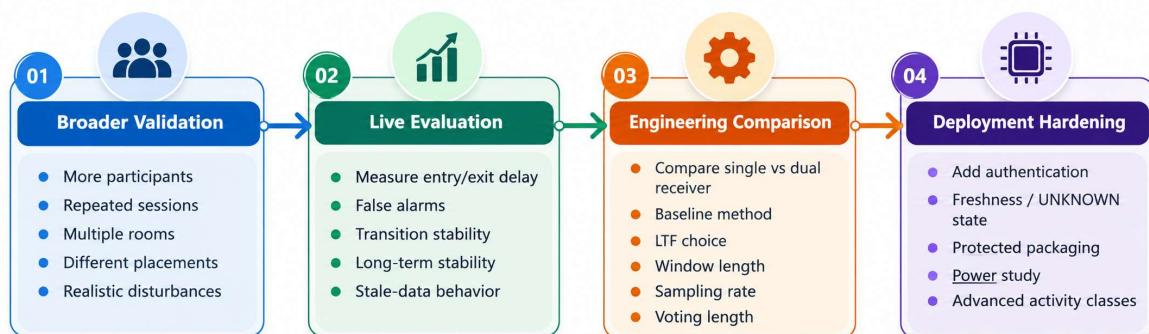
Timeline of Project Implementation

Activity	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Project Assignment	■											
Literature Review, Paper Study and Planning		■	■									
Establish Overall System Architecture			■	■								
ESP-IDF Firmware and CSI Acquisition				■	■	■						
Dual-RX Data Collection and Preprocessing							■	■				
Presence and Activity Model Development								■	■	■		
System Integration and Testing										■	■	
Documentation and Demo Preparation											■	
Final Presentation												■



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Future Work



Future work will focus on improving system robustness, real-time reliability, scalability, and deployment readiness through broader validation and engineering optimization.



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References

- i. F. Wang, S. Zhou, S. Panev, J. Han, and D. Huang, “**Person-in-WiFi: Fine-Grained Person Perception Using WiFi**,” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 5452–5461. [Open Access The CVF](#)
- ii. K. Yan, F. Wang, B. Qian, H. Ding, J. Han, and X. Wei, “**Person-in-WiFi 3D: End-to-End Multi-Person 3D Pose Estimation with Wi-Fi**,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 969–978. [Open Access The CVF](#)
- iii. Espressif Systems, “**ESP32-S3 Wi-Fi Driver – Channel State Information (CSI)**,” *ESP-IDF Programming Guide*. This is relevant for CSI collection using the ESP32-S3 and APIs such as `esp_wifi_set_csi_config()` and `esp_wifi_set_csi()`. [Espressif Systems](#)
- iv. Google, “**MediaPipe Pose Landmarker**,” *Google AI Edge Documentation*. This is relevant to your initial failed attempt where video frames were converted into body landmarks/3D pose ground truth.



Thank You

